Chapter 3 Assignment Due date: May 11, 2026 (Mon)

黃丞暘 611211106

3.1

Sol.

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu \neq \mu_0$$

Under H_0,

$$\bar{X}_i \sim N\left(\mu_0, \frac{\sigma^2}{m}\right)$$

Hence,

$$Z_i = \frac{\bar{X}_i - \mu_0}{\sigma/\sqrt{m}} \sim N(0,1)$$

For a two-sided test with significance level α,

$$P_{H_0}\left(|Z_i| > z_{1-\frac{\alpha}{2}}\right) = \alpha$$

Therefore, the rejection region is

$$|Z_i| > z_{1-\frac{\alpha}{2}}$$

$$\Rightarrow \left|\frac{\bar{X}_i - \mu_0}{\sigma/\sqrt{m}}\right| > z_{1-\frac{\alpha}{2}}$$

$$\Rightarrow \frac{\bar{X}_i - \mu_0}{\sigma/\sqrt{m}} > z_{1-\frac{\alpha}{2}} \quad \text{or} \quad \frac{\bar{X}_i - \mu_0}{\sigma/\sqrt{m}} < -z_{1-\frac{\alpha}{2}}$$

Since $\sigma/\sqrt{m} > 0$,

$$\bar{X}_i - \mu_0 > z_{1-\frac{\alpha}{2}}\frac{\sigma}{\sqrt{m}} \quad \text{or} \quad \bar{X}_i - \mu_0 < -z_{1-\frac{\alpha}{2}}\frac{\sigma}{\sqrt{m}}$$

$$\Rightarrow \bar{X}_i > \mu_0 + z_{1-\frac{\alpha}{2}}\frac{\sigma}{\sqrt{m}} \quad \text{or} \quad \bar{X}_i < \mu_0 - z_{1-\frac{\alpha}{2}}\frac{\sigma}{\sqrt{m}}$$

Let

$$U = \mu_0 + z_{1-\frac{\alpha}{2}}\frac{\sigma}{\sqrt{m}}, \qquad L = \mu_0 - z_{1-\frac{\alpha}{2}}\frac{\sigma}{\sqrt{m}}$$

Therefore,

$$|Z_i| > z_{1-\frac{\alpha}{2}} \iff \text{Reject } H_0 \iff \bar{X}_i < L \quad \text{or} \quad \bar{X}_i > U$$

3.3

Sol.

(i)

Assume

$$\mu_i = \begin{cases} \mu_0, & i < \tau \\ \mu_1, & i \geq \tau \end{cases}$$

where

$$\mu_1 = \mu_0 + \delta, \qquad \delta \neq 0$$

From equation (3.4),

$$\hat{\mu}_0 = \bar{\bar{X}} = \frac{1}{n}\sum_{i=1}^{n} \bar{X}_i$$

Since there are $\tau - 1$ subgroups with mean μ_0, and $n - \tau + 1$ subgroups with mean μ_1,

$$E(\bar{X}_i) = \begin{cases} \mu_0, & i < \tau \\ \mu_1, & i \geq \tau \end{cases}$$

Therefore,

$$\begin{aligned} E(\hat{\mu}_0) &= E\left(\frac{1}{n}\sum_{i=1}^{n} \bar{X}_i\right) = \frac{1}{n}\sum_{i=1}^{n} E(\bar{X}_i) = \frac{1}{n}\left[(\tau - 1)\mu_0 + (n - \tau + 1)\mu_1\right] \\ &= \frac{1}{n}\left[(\tau - 1)\mu_0 + (n - \tau + 1)(\mu_0 + \delta)\right] = \frac{1}{n}\left[n\mu_0 + (n - \tau + 1)\delta\right] \\ &= \mu_0 + \frac{n - \tau + 1}{n}\delta \end{aligned}$$

Thus,

$$\text{Bias}(\hat{\mu}_0) = E(\hat{\mu}_0) - \mu_0 = \frac{n - \tau + 1}{n}\delta$$

Since

$$\delta \neq 0 \quad \text{and} \quad 1 < \tau \leq n$$

we have

$$\frac{n - \tau + 1}{n}\delta \neq 0 \Rightarrow E(\hat{\mu}_0) \neq \mu_0$$

Therefore,

$$\hat{\mu}_0 \text{ is not an unbiased estimator of } \mu_0$$

(ii)

For the sampling distribution, assume

$$X_{ij} \sim N(\mu_i, \sigma^2)$$

Then

$$\bar{X}_i \sim N\left(\mu_i, \frac{\sigma^2}{m}\right)$$

Since $\hat{\mu}_0 = \frac{1}{n}\sum_{i=1}^{n} \bar{X}_i$ and $\bar{X}_1, \ldots, \bar{X}_n$ are independent normal random variables,

$$\hat{\mu}_0 \sim N\left(E(\hat{\mu}_0), Var(\hat{\mu}_0)\right)$$

Compute

$$Var(\hat{\mu}_0) = Var\left(\frac{1}{n}\sum_{i=1}^{n}\bar{X}_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}Var(\bar{X}_i) = \frac{1}{n^2}\sum_{i=1}^{n}\frac{\sigma^2}{m} = \frac{\sigma^2}{mn}$$

Therefore,

$$\hat{\mu}_0 \sim N\left(\mu_0 + \frac{n-\tau+1}{n}\delta, \frac{\sigma^2}{mn}\right)$$

3.5

Sol.

$$n = 10, \quad m = 5, \quad \alpha = 0.0027 \Rightarrow z_{1-\frac{\alpha}{2}} \approx 3$$

(i)

$$\bar{\bar{X}} = \frac{1}{10}\sum_{i=1}^{10}\bar{X}_i = \frac{35.53+36.01+34.74+35.30+35.80+34.41+35.35+35.19+35.05+35.66}{10}$$
$$= 35.304$$

$$\bar{R} = \frac{1}{10}\sum_{i=1}^{10}R_i = \frac{2.41+3.24+2.24+3.16+1.06+1.80+1.60+1.42+1.77+1.42}{10}$$
$$= 2.012$$

For subgroup size $m = 5$,

$$d_1(5) = 2.326, \qquad d_2(5) = 0.864$$

The $\bar{X}$ chart,

$$U_{\bar{X}} = \bar{\bar{X}} + \frac{z_{1-\frac{\alpha}{2}}}{d_1(m)\sqrt{m}}\bar{R} = 35.304 + \frac{3}{2.326\sqrt{5}}\cdot 2.012 \approx 36.465$$

$$C_{\bar{X}} = \bar{\bar{X}} = 35.304$$

$$L_{\bar{X}} = \bar{\bar{X}} - \frac{z_{1-\frac{\alpha}{2}}}{d_1(m)\sqrt{m}}\bar{R} = 35.304 - \frac{3}{2.326\sqrt{5}}\cdot 2.012 \approx 34.143$$

Hence,

$$34.143 < \bar{X}_i < 36.465, \qquad i = 1, 2, \ldots, 10$$

The R chart,

$$U_R = \left(1 + \frac{z_{1-\frac{\alpha}{2}}d_2(m)}{d_1(m)}\right)\bar{R} = \left(1 + \frac{3\cdot 0.864}{2.326}\right)\cdot 2.012 \approx 4.254$$

$$C_R = \bar{R} = 2.012$$

$$L_R = \left(1 - \frac{z_{1-\frac{\alpha}{2}}d_2(m)}{d_1(m)}\right)\bar{R} = \left(1 - \frac{3\cdot 0.864}{2.326}\right)\cdot 2.012 \approx -0.23 \Rightarrow L_R = 0$$

Hence,

$$0 < R_i < 4.254, \qquad i = 1, 2, \ldots, 10$$

In conclusion, the process seems to be in statistical control.

(ii)

The IC mean is estimated by

$$\hat{\mu}_0 = \bar{\bar{X}} = 35.304$$

The IC standard deviation is estimated by

$$\hat{\sigma} = \frac{\bar{R}}{d_1(m)} = \frac{2.012}{2.326} = 0.865$$

Therefore,

$$\hat{\mu}_0 = 35.304, \quad \hat{\sigma} = 0.865$$

(iii)

When the true process mean is not shifted,

$$\bar{X} \sim N\left(\hat{\mu}_0, \frac{\hat{\sigma}^2}{m}\right)$$

The $\bar{X}$ chart gives a signal if

$$\bar{X} < L_{\bar{X}} \quad \text{or} \quad \bar{X} > U_{\bar{X}}$$

Thus,

$$\begin{aligned}
P(\text{signal}) &= P\left(\bar{X} < L_{\bar{X}}\right) + P\left(\bar{X} > U_{\bar{X}}\right) \\
&= P\left(Z < \frac{L_{\bar{X}} - \hat{\mu}_0}{\hat{\sigma}/\sqrt{m}}\right) + P\left(Z > \frac{U_{\bar{X}} - \hat{\mu}_0}{\hat{\sigma}/\sqrt{m}}\right) \\
&= P(Z < -3) + P(Z > 3) = 2[1 - \Phi(3)] = 0.0027
\end{aligned}$$

(iv)

Under the IC process,

$$ARL_0 = \frac{1}{\alpha} = \frac{1}{0.0027} = 370.37$$

For a mean shift of size 1,

$$\mu_1 = \mu_0 + k\sigma = \mu_0 + 1 \Rightarrow k = \frac{\mu_1 - \mu_0}{\hat{\sigma}} = \frac{1}{0.865} = 1.156$$

Then

$$\bar{X} \sim N\left(\mu_0 + 1, \frac{\hat{\sigma}^2}{m}\right)$$

The probability of no signal after the shift is

$$\begin{aligned}
\beta &= P\left(\mu_0 - z_{1-\frac{\alpha}{2}}\frac{\sigma}{\sqrt{m}} < \bar{X} < \mu_0 + z_{1-\frac{\alpha}{2}}\frac{\sigma}{\sqrt{m}}\right) \\
&= P\left(-k\sqrt{m} - z_{1-\frac{\alpha}{2}} < \frac{\bar{X} - \mu_1}{\sigma/\sqrt{m}} < -k\sqrt{m} + z_{1-\frac{\alpha}{2}}\right) \\
&= \Phi\left(-k\sqrt{m} + z_{1-\frac{\alpha}{2}}\right) - \Phi\left(-k\sqrt{m} - z_{1-\frac{\alpha}{2}}\right) \\
&= \Phi\left(-1.156\sqrt{5} + 3\right) - \Phi\left(-1.156\sqrt{5} - 3\right) \\
&= \Phi(0.415) - \Phi(-5.585) \approx 0.6609
\end{aligned}$$

Therefore,

$$ARL_1 = \frac{1}{1-\beta} = \frac{1}{1-0.6609} = 2.949$$

3.6

Sol.
From Exercise 3.5,

$$\bar{\bar{X}} = 35.304 \quad \text{and} \quad \bar{R} = 2.012$$

$$\bar{s} = \frac{1}{10}\sum_{i=1}^{10} s_i = \frac{0.99 + 1.28 + 0.97 + 1.25 + 0.43 + 0.70 + 0.77 + 0.53 + 0.66 + 0.63}{10} = 0.821$$

For subgroup size $m = 5$,

$$c_4(5) = 0.9400, \quad A_3 = 1.427, \quad B_3 = 0, \quad B_4 = 2.089, \quad d_1(5) = 2.326, \quad d_2(5) = 0.864$$

(i)

The $\bar{X}$ chart based on s

$$U_{\bar{X}} = \bar{\bar{X}} + A_3\bar{s} = 35.304 + 1.427 \cdot 0.821 \approx 36.476$$

$$C_{\bar{X}} = \bar{\bar{X}} = 35.304$$

$$L_{\bar{X}} = \bar{\bar{X}} - A_3\bar{s} = 35.304 - 1.427 \cdot 0.821 \approx 34.132$$

Hence,

$$34.132 < \bar{X}\text{-}s < 36.476$$

The s chart,

$$U_s = B_4\bar{s} = 2.089(0.821) = 1.715$$

$$C_s = \bar{s} = 0.821$$

$$L_s = B_3\bar{s} = 0(0.821) = 0$$

Hence,

$$0 < s_i < 1.715$$

In conclusion, the process seems to be in statistical control.

(ii)

The $\bar{X}$ chart based on R,

$$U_{\bar{X},R} = \bar{\bar{X}} + \frac{3}{d_1(5)\sqrt{5}}\bar{R} = 35.304 + \frac{3}{2.326\sqrt{5}}(2.012) = 36.465$$

$$L_{\bar{X},R} = \bar{\bar{X}} - \frac{3}{d_1(5)\sqrt{5}}\bar{R} = 35.304 - \frac{3}{2.326\sqrt{5}}(2.012) = 34.144$$

Hence,

$$34.144 < \bar{X} < 36.465$$

The R chart,

$$U_R = \left(1 + 3\frac{d_2(5)}{d_1(5)}\right)\bar{R} = \left(1 + 3\frac{0.864}{2.326}\right)(2.012) = 4.254$$

$$C_R = \bar{R} = 2.012$$

$$L_R = \left(1 - 3\frac{d_2(5)}{d_1(5)}\right)\bar{R} = \left(1 - 3\frac{0.864}{2.326}\right)(2.012) = -0.229 \Rightarrow D_3 = 0$$

Hence,

$$0 < R < 4.254$$

The two $\bar{X}$ charts have limits

$$\bar{X}\text{-}s: \quad [34.132,\ 36.476]$$

$$\bar{X}\text{-}R: \quad [34.144,\ 36.465]$$

These limits are almost identical.
Both the s chart and the R chart show no out-of-control signal.
Both methods suggest that the process is in statistical control

(iii)

Estimators of σ
The range-based estimator is

$$\hat{\sigma}_R = \frac{\bar{R}}{d_1(m)} = \frac{2.012}{2.326} = 0.8650$$

The sample-standard-deviation-based estimator is

$$\hat{\sigma}_s = \frac{\bar{s}}{c_4(m)} = \frac{0.821}{0.9400} = 0.8734$$

The combined-sample standard deviation is

$$\hat{\sigma}_{\text{combined}} = s_{\text{combined}} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{10}\sum_{j=1}^{5}\left(X_{ij} - \bar{X}_{\text{all}}\right)^2} = 0.9099$$

The range-based estimator is simple, but it uses only the minimum and maximum values in each subgroup. The s-based estimator uses all observations inside each subgroup and is usually more efficient. The combined-sample estimator uses all observations, but it mixes within-subgroup and between-subgroup variation; therefore, it may overestimate the natural process variation when subgroup means differ over time. For SPC, $\hat{\sigma}_R$ and $\hat{\sigma}_s$ are more appropriate for estimating within-subgroup variation

3.7

Sol.

i.

3.8

Sol.

i.

3.9

Sol.

i.

3.11

Sol.

i.

3.13

Sol.

i.

3.20

Sol.

i.